

Practice Midterm #2

Name: _____

ID: _____

Score: _____

Problem	Pts.	Score
1	20	
2	20	
3	20	
Total	60	

I certify that this is my sole work and I haven't discussed any problem with other person.

Signature: _____

1. Please print your name and student ID number.
2. Sign on the above statement.
3. For full credit, you should show all of your work and explain your reasoning.

GOOD LUCK!

Problem 1 (20 Pts) (a) (10 Pts) Let $h[n]$ be an FIR filter of GLP type I, i.e., the length L of the filter is an odd number and the impulse response shows the symmetry around the mid-point. Let $g[n]$ be obtained through its z -transform satisfying

$$G(z) = -z^{-(L-1)}H(-z^{-1})$$

where $H(z)$ denotes the z -transform of $h[n]$. Determine whether $g[n]$ has GLP or not.

(b) (10 Pts) Consider a discrete-time filter $h[n]$ whose z -transform is given by

$$H(z) = z^{-4}(z - 0.5e^{j\pi/4})(z + 0.5e^{j\pi/4})(z - 2e^{j\pi/4})(z + 2e^{j\pi/4}).$$

Determine whether $h[n]$ has GLP or not.

Problem 2 (20 Pts) Suppose that we are given a continuous-time lowpass filter with frequency response $H_c(j\Omega)$ such that

$$\begin{aligned} 1 - \delta_p \leq |H_c(j\Omega)| \leq 1 + \delta_p, & \quad |\Omega| \leq \Omega_p, \\ |H_c(j\Omega)| \leq \delta_s, & \quad |\Omega| \geq \Omega_s. \end{aligned}$$

A set of discrete-time lowpass filters is derived from $H_c(s)$ via the bilinear transformation

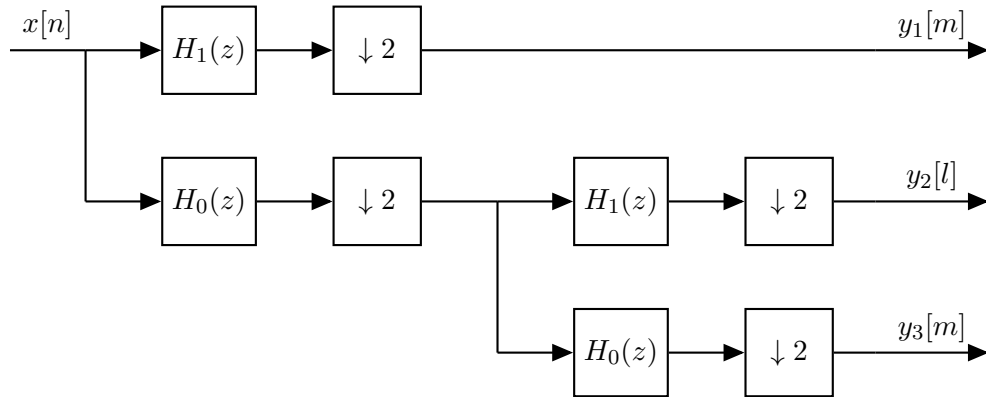
$$H(z) = H_c(s) \Big|_{s = \frac{2}{T} \cdot \frac{1-z^{-1}}{1+z^{-1}}}$$

parameterized by $T > 0$.

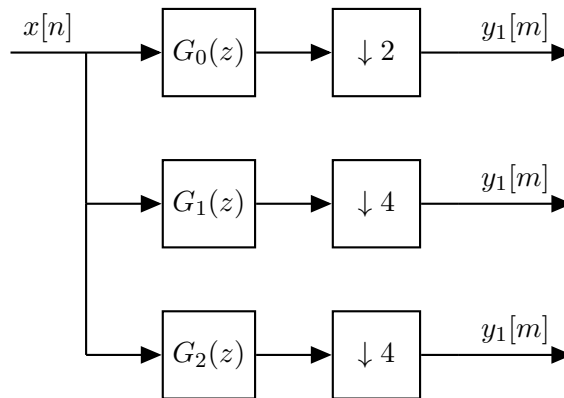
(a) **(10 Pts)** Show that under this transformation, the imaginary axis $s = j\Omega$ in the s -plane is mapped to the unit circle $z = e^{j\omega}$ in the z -plane. Derive the relationship between ω and Ω .

(b) **(10 Pts)** Given a fixed Ω_p , find the value of T such that the corresponding cutoff frequency for the discrete-time system is $\omega_p = \pi/2$.

Problem 3 (20 Pts) Consider the following filterbank system with $H_0(z) = 1 + z^{-1}$ and $H_1(z) = 1 - z^{-1}$:



- (a) (10 Pts) Find $G_0(z)$, $G_1(z)$, $G_2(z)$, expressed in terms of $H_0(z)$ and $H_1(z)$, such that the following system produces the same outputs.



(b) (10 Pts) Sketch the magnitude responses of $G_0(e^{j\omega})$, $G_1(e^{j\omega})$, and $G_2(e^{j\omega})$.